

Project Design Phase-II Data Flow Diagram & User Stories

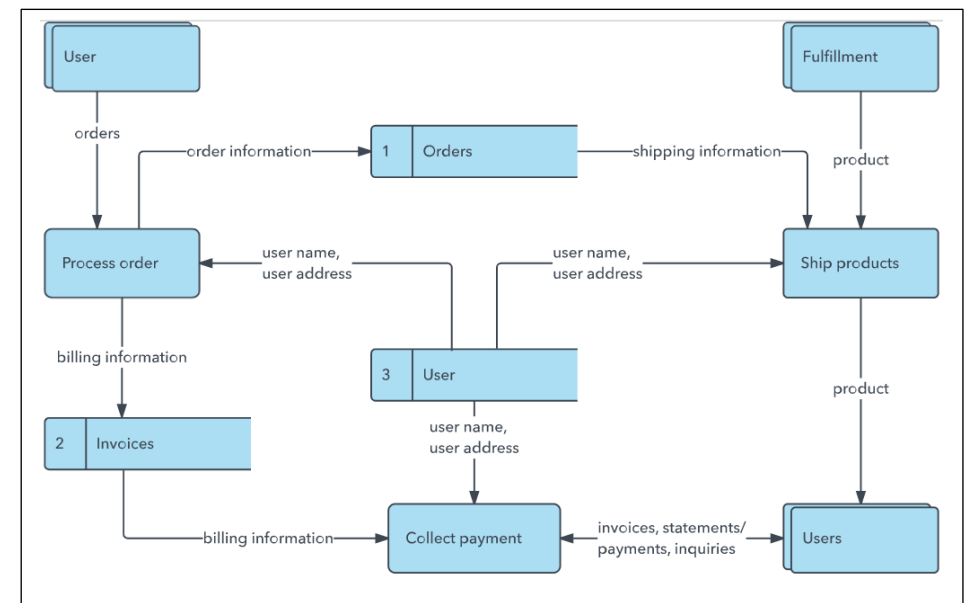
Date	30 September 2026
Team ID	SWTID-2026-8629
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Data Flow Diagrams:

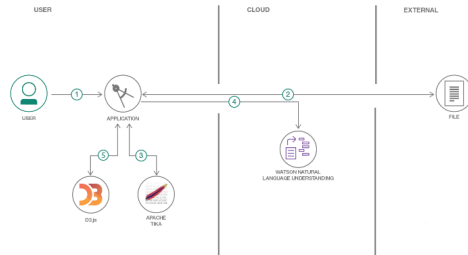
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [\(Simplified\)](#)

Example: DFD Level 0 (Industry Standard)



Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Requester	Catalog Request	USN-1	As a requester, I can submit the Standard Laptop catalog item.	Item is active and request is submitted.	High	Sprint-1
Requester	Request Tracking	USN-2	As a requester, I can receive a request and RITM.	REQ/RITM are created.	High	Sprint-1
Administrator	Flow Automation	USN-3	As an administrator, I can configure the Service Catalog trigger.	Flow starts from catalog request.	High	Sprint-1
Administrator	Task Creation	USN-4	As an administrator, I can configure Create Catalog Task.	SCTASK is generated.	High	Sprint-1
Fulfillment User	Task Processing	USN-5	As a fulfillment user, I can receive a structured task.	Task is visible with reference.	High	Sprint-1
Administrator	Validation	USN-6	As an administrator, I can verify REQ → RITM → SCTASK.	Chain is traceable.	High	Sprint-1
IT Manager	Scalability	USN-7	As an IT manager, I can reuse the pattern for other IT items.	Pattern is reusable.	Medium	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Requester	Usability	USN-8	As a requester, I can use standard ServiceNow UI.	No separate app required.	Medium	Sprint-1
Administrator	Maintenance	USN-9	As an administrator, I can update the flow.	Flow remains configurable.	Medium	Sprint-2
Fulfillment User	Task Status	USN-10	As a fulfillment user, I can update task state.	Task state can progress.	Medium	Sprint-1
Administrator	Access	USN-11	As an administrator, I can restrict configuration access.	Only authorized roles can configure.	Medium	Sprint-1
Requester	Completion	USN-12	As a requester, I can trace fulfillment from my request.	Request links to task.	High	Sprint-1

Completed Data Flow

Project: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Student: Dharshini M

Team: SWTID-2026-8629

Date: 30 September 2026

ServiceNow Standard Laptop Request Automation — Solution Architecture

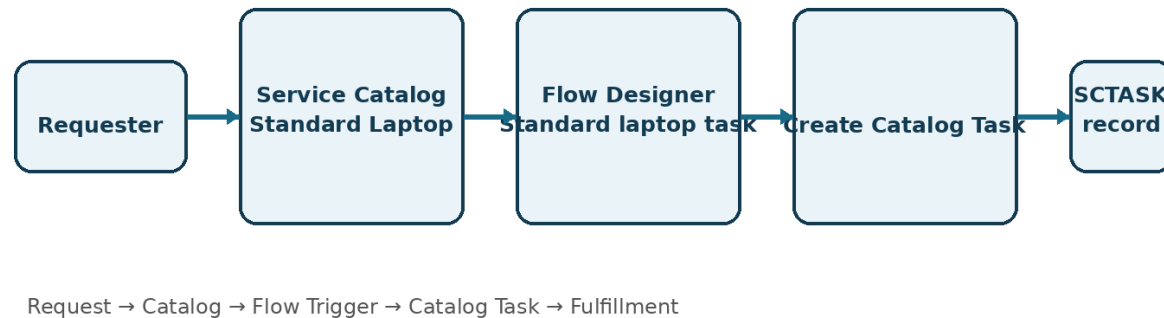


Figure: ServiceNow solution architecture and data flow.

Data flow

Requester → Standard Laptop Catalog Item → Service Catalog Trigger → Standard laptop task flow → Create Catalog Task → SCTASK record